

# Alexandre Champagne-Ruel

NASA Postdoctoral Program Fellow

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*Origins of Life • Astrobiology • Complex Systems*

## RESEARCH PROFILE

Physicist and NASA Postdoctoral Fellow studying why driven chemical systems become complex rather than settle at equilibrium. I use stochastic artificial-chemistry simulations and assembly theory to test how environmental structure and energy flow shape molecular complexity before Darwinian evolution, and I am extending the same questions to biological regulatory networks. The applied aim is an abiotic baseline for agnostic life detection.

## ACADEMIC POSITIONS

**NASA Postdoctoral Program Fellow**  
Arizona State University, Mathis Group

**2025 – Current**  
Tempe, AZ, USA

## EDUCATION

**Ph.D. Physics**

**2025**

Université de Montréal

Montréal, QC, Canada

- Thesis: *Spatiality in Prebiotic Evolution: Toward a Physics of the Emergence of Complexity*
- Advisor: Paul Charbonneau

**M.Sc. Physics**

**2020**

Université de Montréal

Montréal, QC, Canada

- Thesis: *From Game Theory to Exobiology: The Emergence of Cooperation as a Critical Phenomenon*
- Advisor: Paul Charbonneau

**B.Sc. Physics**

**2018**

Université de Montréal

Montréal, QC, Canada

**B.Sc. Philosophy**

**2012**

Université de Montréal

Montréal, QC, Canada

## GRANTS & AWARDS

*Only individual grants >\$5,000 are listed.*

**NASA Postdoctoral Program Fellowship** NASA

146,496 USD

**2025**

*Project: Mapping Molecular Complexity for Agnostic Life Detection*

**J. Armand Bombardier Scholarship** Fondation Bombardier

10,000 CAD

**2024**

**Globalink Research Award** Mitacs Canada

6,000 CAD

**2024**

**Google Cloud Research Grant** Google

5,000 USD

**2023 – 2026**

**J. Armand Bombardier Scholarship** Fondation Bombardier

10,000 CAD

**2023**

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|---------------------------------------------------------------------|-------------|--------------------|
| <b>J. Armand Bombardier Scholarship</b> <i>Fondation Bombardier</i> | 10,000 CAD  | <b>2022</b>        |
| <b>Doctoral Scholarship</b> <i>Fonds de Recherche du Québec</i>     | 70,000 CAD  | <b>2022</b>        |
| <b>Excellence Award</b> <i>Université de Montréal</i>               | 10,000 CAD  | <b>2018</b>        |
| <b>Various Travel &amp; Research Grants</b> <i>15 awards</i>        | ~30,000 CAD | <b>2011 – 2025</b> |

## PUBLICATIONS

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### Accepted Articles

- A. **Champagne-Ruel**, C. P. Kempes, and C. Mathis, “Spatial Patterning and Selection: How the Environment Shapes Molecular Complexity”, [10.48550/arXiv.2509.04547](https://doi.org/10.48550/arXiv.2509.04547) (2025), (Accepted for publication in PNAS Nexus).

### Published Articles

- S. Asche, C. Bautista, C. Blanco, D. Boulesteix, A. **Champagne-Ruel**, C. Mathis, O. Markovitch, et al., “What it takes to solve the origin of life: An integrated review. Part 1–Experimental methods and data repositories”, [10.1016/j.xcrp.2026.103212](https://doi.org/10.1016/j.xcrp.2026.103212) (2026).
- S. Asche, C. Bautista, C. Blanco, D. Boulesteix, A. **Champagne-Ruel**, C. Mathis, O. Markovitch, et al., “What it takes to solve the origins of life: An integrated review. Part 2: Theoretical methods and emerging trends”, [10.1016/j.xcrp.2026.103211](https://doi.org/10.1016/j.xcrp.2026.103211) (2026).
- S. Rajamani, C. Mathis, S. Asche, D. Boulesteix, C. Blanco, O. Markovitch, A. **Champagne-Ruel**, et al., “Better origin(s) of life research through collaboration across disciplines”, [10.1016/j.xcrp.2026.103220](https://doi.org/10.1016/j.xcrp.2026.103220) (2026).
- A. **Champagne-Ruel**, S. Zakaib-Bernier, and P. Charbonneau, “Diffusion and pattern formation in spatial games”, *Physical Review E* **110**, 014301 (2024).
- A. **Champagne-Ruel** and P. Charbonneau, “A Mutation Threshold for Cooperative Takeover”, *Life* **12**, 254 (2022).
- S. Gelin, A. **Champagne-Ruel**, and N. Mousseau, “Enthalpy-entropy compensation of atomic diffusion originates from softening of low frequency phonons”, *Nature Communications* **11**, 3977 (2020).

## PRESENTATIONS

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### Invited Talks

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|------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Spatiality in Prebiotic Evolution</b><br><i>Queens College, City University of New York</i>                   | <b>2026</b><br><i>New York, USA</i>        |
| <b>Physics of Complexity And Agnostic Life Detection</b><br><i>Trottier Institute for Research on Exoplanets</i> | <b>2025</b><br><i>Montréal, QC, Canada</i> |
| <b>Spatial Patterning and Selection</b><br><i>Max Planck Institute for Terrestrial Microbiology</i>              | <b>2025</b><br><i>Marburg, Germany</i>     |
| <b>From Emergent Complexity to Reliable Life Detection</b><br><i>Arizona State University</i>                    | <b>2024</b><br><i>Tempe, AZ, USA</i>       |
| <b>Coopération, émergence et transitions</b><br><i>Université du Québec à Montréal</i>                           | <b>2022</b><br><i>Montréal, QC, Canada</i> |

## Contributed Talks

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|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>How Environments Shape Molecular Complexity: From Abiogenesis to Life Detection</b><br><i>Interdisciplinary Origin of Life Meeting (IOoL)</i>      | <b>2026</b><br><i>Tokyo, Japan</i>             |
| <b>Network Topology as a Tunable Parameter for Emergent Complexity</b><br><i>Conference on Complex Systems (CCS)</i>                                  | <b>2026</b><br><i>Binghamton, NY, USA</i>      |
| <b>From Emergent Complexity to Reliable Life Detection</b><br><i>Biennial European Astrobiology Conference (BEACON)</i>                               | <b>2025</b><br><i>Reykjavik, Iceland</i>       |
| <b>Diffusion: An Overlooked Driver of Prebiotic Complexity</b><br><i>Astrobiology Science Conference (AbSciCon)</i>                                   | <b>2024</b><br><i>Providence, RI, USA</i>      |
| <b>Théorie de l'information et origine de la vie</b><br><i>90e Congrès de l'ACFAS</i>                                                                 | <b>2023</b><br><i>Montréal, QC, Canada</i>     |
| <b>A Mutation Threshold for Cooperative Takeover</b><br><i>Astrobiology Science Conference (AbSciCon)</i>                                             | <b>2022</b><br><i>Atlanta, GA, USA</i>         |
| <b>Cooperation: An Emergent Universal Feature at the Dawn of Life</b><br><i>Interdisciplinary Origin of Life Meeting for Early Career Researchers</i> | <b>2022</b><br><i>Montréal, QC, Canada</i>     |
| <b>Mutation Favors the Emergence of Cooperation</b><br><i>Life and Space Poland</i>                                                                   | <b>2021</b><br><i>Virtual/Szczecin, Poland</i> |
| <b>La criticité dans un système évolutif artificiel</b><br><i>Centre de recherche en astrophysique du Québec (CRAQ)</i>                               | <b>2019</b><br><i>Orford, QC, Canada</i>       |

## Posters

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|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>How Environments Shape Molecular Complexity: From Abiogenesis to Life Detection</b><br><i>Astrobiology Science Conference (AbSciCon)</i>      | <b>2026</b><br><i>Madison, WI, USA</i>        |
| <b>Mapping Molecular Complexity for Agnostic Life Detection</b><br><i>NASA Postdoctoral Program Virtual Symposium</i>                            | <b>2025</b><br><i>Virtual</i>                 |
| <b>L'émergence de la coopération via l'évolution de réseaux informationnels</b><br><i>90e Congrès de l'ACFAS</i>                                 | <b>2023</b><br><i>Montréal, QC, Canada</i>    |
| <b>Diffusion, structures spatiales et origine de la vie</b><br><i>90e Congrès de l'ACFAS</i>                                                     | <b>2023</b><br><i>Montréal, QC, Canada</i>    |
| <b>OoLEN - The Origin of Life Early-career Network: Building the Community Needed to Solve the Problem</b><br><i>AbSciCon</i>                    | <b>2022</b><br><i>Atlanta, GA, USA</i>        |
| <b>A Mutation Threshold for Cooperative Takeover</b><br><i>Gordon Research Seminar: Challenging Paradigms in Prebiotic Chemistry</i>             | <b>2022</b><br><i>Oxnard, CA, USA</i>         |
| <b>A Mutation Threshold for Cooperative Takeover</b><br><i>Gordon Research Conference: Environments for the Origins of Life and Habitability</i> | <b>2022</b><br><i>Oxnard, CA, USA</i>         |
| <b>Les mutations favorisent la coopération en contexte évolutif</b><br><i>Centenaire, Département de Physique, Université de Montréal</i>        | <b>2021</b><br><i>Montréal, QC, Canada</i>    |
| <b>Mutation Favors the Emergence of Cooperative Behavior</b><br><i>Molecular Origins of Life Munich</i>                                          | <b>2021</b><br><i>Virtual/Munich, Germany</i> |

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|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>Mutations Promote Cooperation in an Evolutionary Setting</b><br><i>XIXth ISSOL Conference</i>                      | <b>2021</b><br><i>Virtual</i>            |
| <b>Self-Organized Criticality : A Prelude to Avalanche Models of Solar Flares</b><br><i>Space Climate 7 Symposium</i> | <b>2019</b><br><i>Orford, QC, Canada</i> |

## SCHOOLS & SELECTIVE WORKSHOPS

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|------------------------------------------------------------------------------|------------------------------------------|
| <b>Assembly Theory for Folded Matter</b><br><i>Santa Fe Institute</i>        | <b>2025</b><br><i>Santa Fe, NM, USA</i>  |
| <b>Information Driven States of Matter</b><br><i>University of Rochester</i> | <b>2024</b><br><i>Rochester, NY, USA</i> |

## TEACHING EXPERIENCE

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| <b>Guest Lecturer – CAS 201: Scientific Programming with Python</b><br><i>Arizona State University</i>                                                        | <b>Fall 2026</b><br><i>Tempe, AZ, USA</i>         |
| <b>Mentor – Biocomputing Scholars program</b><br><i>Arizona State University</i>                                                                              | <b>2025 – 2026</b><br><i>Tempe, AZ, USA</i>       |
| <ul style="list-style-type: none"> <li>Guiding 3 undergraduate students in developing computational chemistry simulations.</li> </ul>                         |                                                   |
| <b>Undergraduate Internship Supervision</b><br><i>Université de Montréal</i>                                                                                  | <b>2022</b><br><i>Montréal, QC, Canada</i>        |
| <ul style="list-style-type: none"> <li>Supervised a summer research project on diffusion; resulted in journal publication and poster presentation.</li> </ul> |                                                   |
| <b>Teaching Assistant – Introduction to Astrobiology</b><br><i>Université de Montréal</i>                                                                     | <b>2021 – 2022</b><br><i>Montréal, QC, Canada</i> |
| <b>Tutoring – Undergraduate Level</b><br><i>Université de Montréal</i>                                                                                        | <b>2018 – 2022</b><br><i>Montréal, QC, Canada</i> |

## SERVICE & OUTREACH

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### Scientific Leadership

#### Positions

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Member of the Executive Board</b><br><i>Origin of Life Early-career Network (OoLEN)</i>                                                                            | <b>2022 – Present</b> |
| <ul style="list-style-type: none"> <li>Previously head of the virtual-meetings workgroup; organizer of the OoLEN Public Seminars lecture series (2023–24).</li> </ul> |                       |
| <b>Founder and Convener</b><br><i>cs-astrobio, international journal club and discussion group on complex-systems methods in astrobiology</i>                         | <b>2026 – Present</b> |
| <b>Origin of Life Digest (<a href="#">link</a>)</b><br><i>Editor</i>                                                                                                  | <b>2021 – Present</b> |

#### Conference Organization

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|-------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Interdisciplinary Origin of Life Meeting</b><br><i>Organizing Committee</i>                        | <b>2026</b><br><i>Tokyo, Japan</i>         |
| <b>Frontiers in Astrobiology and Origins of Life Conference</b><br><i>Organizing Committee</i>        | <b>2025</b><br><i>Reykjavik, Iceland</i>   |
| <b>Origine de la vie : de l'astrophysique à la philosophie</b><br><i>Lead Organizer</i>               | <b>2023</b><br><i>Montréal, QC, Canada</i> |
| <b>Interdisciplinary Origin of Life Meeting for Early Career Researchers</b><br><i>Lead Organizer</i> | <b>2022</b><br><i>Montréal, QC, Canada</i> |
| <b>Space Climate 7</b><br><i>Local Organizing Committee</i>                                           | <b>2019</b><br><i>Orford, QC, Canada</i>   |

## Session Convener

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|-------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>AbSciCon</b><br><i>Assembly Theory Across Scales: From Molecules to Planetary Systems</i>    | <b>2026</b><br><i>Madison, WI, USA</i> |
| <b>AbSciCon</b><br><i>Exploring Self-Assembly and Self-Organization in Prebiotic Complexity</i> | <b>2026</b><br><i>Madison, WI, USA</i> |

## Media & Public Engagement

- É. Beaudoin-Paul, “Entrevue avec Alexandre Champagne-Ruel, récipiendaire de la prestigieuse bourse en astrobiologie de la NASA”, [Quartier Libre \(2025\)](#).
- A. Riopel, “Comment reconnaître la vie sur d’autres planètes”, [Le Devoir \(2023\)](#).
- C. Tamura, “Cooperation and the Origin of Life”, Quantum Photonics Clubhouse Podcast (2022).

## Reviewing Activities

- External Reviewer, NASA Solar System Science Program (2026)
- Journal of the Royal Society Open Science
- Journal of the Royal Society Interface
- Trends in Biochemical Sciences

## Memberships

- International Society for Artificial Life
- Center for Research in Astrophysics of Québec
- Canadian Association of Physicists
- Canadian Astronomical Society
- Origin of Life Early-career Network
- International Society for the Study of the Origin of Life
- Complex Systems Society
- Scientific Society for Astrobiology (Founding Member)

## TECHNICAL SKILLS

- **Languages:** Fluent in Spoken/Written French, English
- **Programming:** Python, C++, Fortran, Julia, R, LaTeX, MATLAB, x86 assembly, Bash
- **Modeling:** Agent-based, Evolutionary Algorithms, Machine Learning, Network Theory, Game Theory
- **Development Workflow:** Linux, High Performance Computing (HPC)/SLURM, Git, Jekyll, Hugo, HTML5/CSS3